Midterm Scratch Work - Biostatistics

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```
# Q7 part a
# Calculate the probability of getting between 3 and 6 successes
# in a binomial distribution with n = 10 trials and p = 0.5 success probability

# P(3 ≤ X ≤ 6) = P(X ≤ 6) - P(X ≤ 2)
pbinom(6, size = 10, prob = 0.5) - pbinom(2, size = 10, prob = 0.5)
```

```
[1] 0.7734375
```

Question 3(b)

Sample size calculation for a follow-up trial:
You want to detect a **0.5-unit difference** in mean antibody response with a **standard deviation of 1.5**, **90% power**, and a **two-sided test**.

We use the sample size formula for comparing two independent means:

$$n = \left(\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\Delta/\sigma} \right)^2$$

Where:

- $$\Delta = 0.5$$

  (effect size to detect)

- $$\sigma = 1.5$$

  (standard deviation)

- $$\alpha = 0.05$$

  (significance level for a two-sided test ⇒ 95% confidence)

- $$\text{Power} = 90\% \Rightarrow \beta = 0.10$$

Although the study is powered at **90%**, we still use a **95% confidence level** for hypothesis testing (i.e.,

$$\alpha = 0.05$$

), which is standard practice in clinical trials and reflected in the formula above.

From the standard normal table (using R):

```
# Z-score for 95% confidence level (two-sided test)
qnorm(1 - 0.05 / 2)  # 1.959964
```

```
[1] 1.959964
```

```
# Z-score for 90% power
qnorm(0.90)  # 1.281552
```

```
[1] 1.281552
```

Plugging into the formula:

We substitute the values into the formula for sample size:

$$n = \left(\frac{1.96 + 1.28}{0.5/1.5} \right)^2 = \left(\frac{3.24}{0.333} \right)^2 = (9.72)^2 \approx 94.5$$

Checking work with R

```
# Z-scores
Z_alpha_2 <- qnorm(1 - 0.05 / 2)  # ≈ 1.96
Z_beta <- qnorm(0.90)             # ≈ 1.28

# Inputs
delta <- 0.5
sigma <- 1.5
Z_total <- Z_alpha_2 + Z_beta

# Sample size calculation
n <- (Z_total / (delta / sigma))^2
n  # ≈ 94.5
```

```
[1] 94.56681
```

Answer:
You would need approximately **95 participants per group** to detect a 0.5-unit difference in mean antibody response with **90% power**, using a **two-sided test** at the **0.05 significance level**.

Question 6(b)

Calculate the p-value for the paired t-test:

```r
# Define test scores for each twin
traditional <- c(78, 85, 67, 93, 72, 60, 71, 49, 75)
montessori  <- c(81, 83, 71, 90, 73, 58, 75, 52, 75)

# Run paired t-test
t_test_result <- t.test(traditional, montessori, paired = TRUE)

# Output t-statistic and p-value
t_test_result$statistic  # t-value
```

```
        t 
-0.9665953 
```

```r
t_test_result$p.value    # p-value
```

```
[1] 0.3620544
```

Q6 part d

```r
t_test_result$conf.int
```

```
[1] -3.009509  1.231731
attr(,"conf.level")
[1] 0.95
```

```r
# Required packages
library(ggplot2)


# Extract CI values and degrees of freedom
ci_low <- t_test_result$conf.int[1]
ci_high <- t_test_result$conf.int[2]
mean_diff <- t_test_result$estimate  # mean of the differences
df <- t_test_result$parameter        # degrees of freedom
t_crit <- qt(0.975, df)

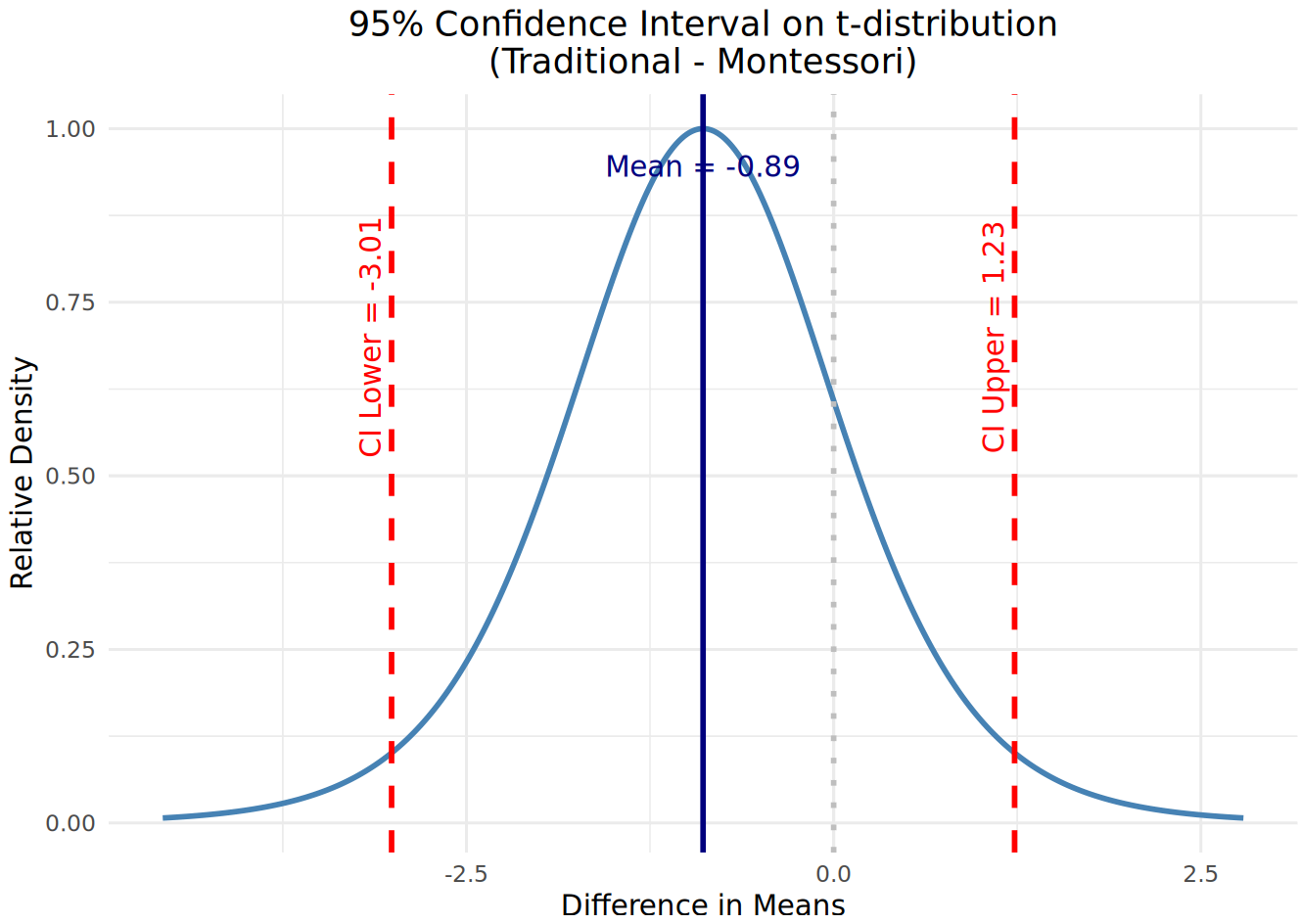
# Estimate standard error from CI width and t critical value
std_err <- (ci_high - mean_diff) / t_crit

# Generate t-distribution
x_vals <- seq(mean_diff - 4*std_err, mean_diff + 4*std_err, length.out = 500)
y_vals <- dt((x_vals - mean_diff) / std_err, df)

# Normalize y for aesthetics
y_vals <- y_vals / max(y_vals)

# Create data frame
plot_data <- data.frame(x = x_vals, y = y_vals)
```

```r
# Plot with labeled CI and mean
ggplot(plot_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_vline(xintercept = mean_diff,
             color = "navy", linetype = "solid", linewidth = 1) +
  geom_vline(xintercept = ci_low,
             color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = ci_high,
             color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = 0,
             color = "gray", linetype = "dotted", linewidth = 1) +
  annotate("text", x = mean_diff,
                   y = 0.9,
                   label = paste0("Mean = ", round(mean_diff, 2)),
                   color = "navy", vjust = -1) +
  annotate("text", x = ci_low,
                   y = 0.7,
                   label = paste0("CI Lower = ", round(ci_low, 2)),
                   color = "red", angle = 90, vjust = -0.5) +
  annotate("text", x = ci_high,
                   y = 0.7,
                   label = paste0("CI Upper = ", round(ci_high, 2)),
                   color = "red", angle = 90, vjust = -0.5) +
  labs(
    title = "95% Confidence Interval on t-distribution\n(Traditional - Montessori)",
    x = "Difference in Means",
    y = "Relative Density"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "none"
  )
```



Question 4

We tested whether mean weight loss differed across the three diet groups using **Welch's ANOVA**, which adjusts for unequal variances between groups.

```
# Load dataset using base R
diet_data <- read.csv("diet.csv")

# Calculate weight loss
diet_data$weight_loss <- diet_data$initial.weight - diet_data$final.weight

# Perform Welch's ANOVA
welch_result <- oneway.test(weight_loss ~ diet.type, data = diet_data, var.equal = FALSE)

# Display the result
welch_result
```

```
    One-way analysis of means (not assuming equal variances)

data:  weight_loss and diet.type
F = 5.2693, num df = 2.00, denom df = 48.48, p-value = 0.008497
```

```r
# Load required package
library(ggplot2)

# Load the dataset
diet_data <- read.csv("diet.csv")

# Calculate weight loss
diet_data$weight_loss <- diet_data$initial.weight - diet_data$final.weight

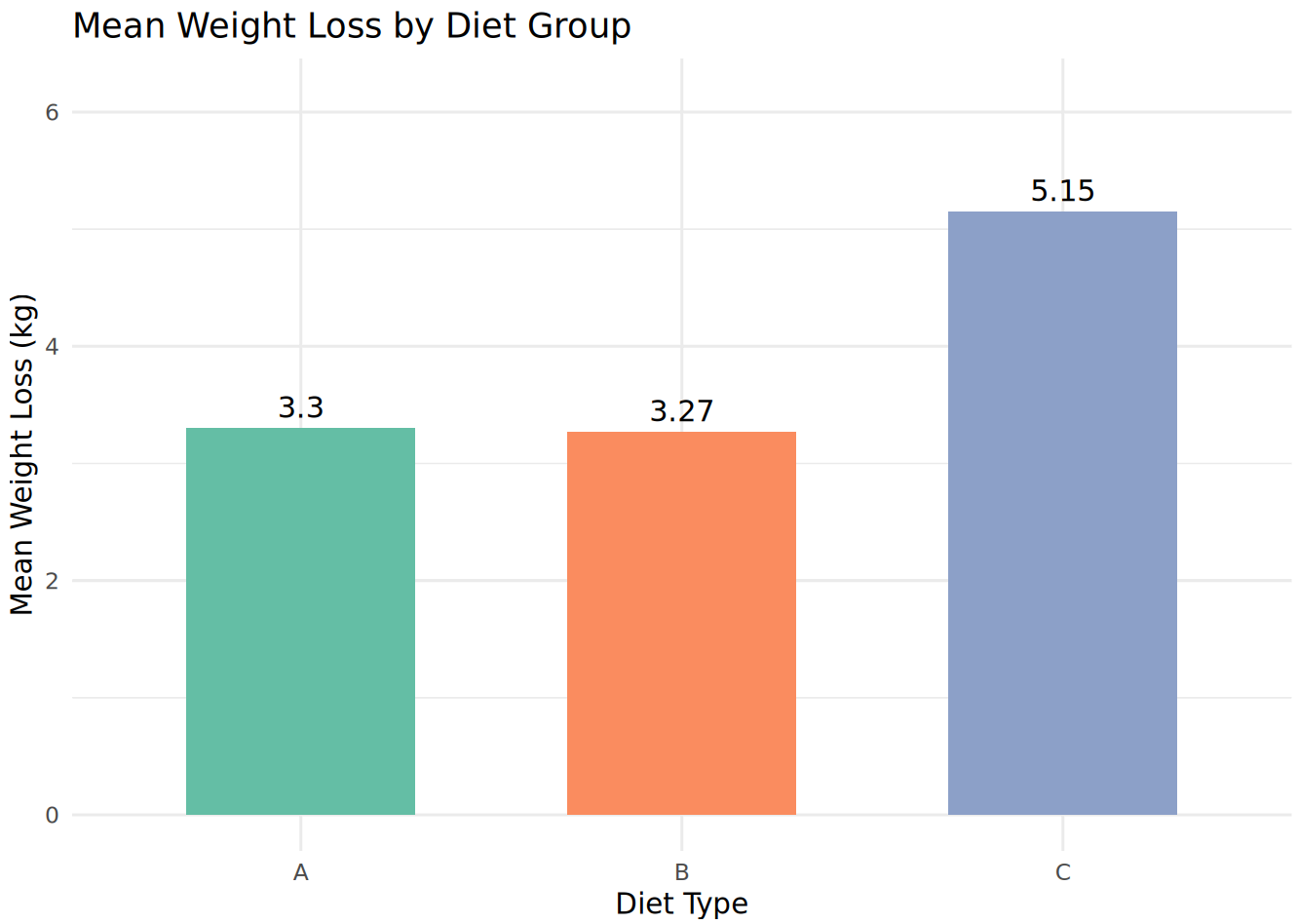
# Calculate mean weight loss by diet group
mean_table <- aggregate(weight_loss ~ diet.type, data = diet_data, FUN = mean)

# Rename columns for clarity
colnames(mean_table) <- c("Diet Type", "Mean Weight Loss (kg)")

# Display the table
mean_table
```

```
  Diet Type Mean Weight Loss (kg)
1         A              3.300000
2         B              3.268000
3         C              5.148148
```

```r
# Plot with colored bars and labels
ggplot(mean_table, aes(x = `Diet Type`, y = `Mean Weight Loss (kg)`, fill = `Diet Type`)) +
  geom_bar(stat = "identity", width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = round(`Mean Weight Loss (kg)`, 2)),
            vjust = -0.5, size = 4, color = "black") +
  scale_fill_manual(values = c("A" = "#66c2a5", "B" = "#fc8d62", "C" = "#8da0cb")) +
  labs(
    title = "Mean Weight Loss by Diet Group",
    x = "Diet Type",
    y = "Mean Weight Loss (kg)"
  ) +
  theme_minimal() +
  ylim(0, max(mean_table$`Mean Weight Loss (kg)`) + 1)
```




```
# Run pairwise t-tests with Holm correction (Welch's method)
pairwise_results <- pairwise.t.test(diet_data$weight_loss,
                                    diet_data$diet.type,
                                    p.adjust.method = "holm",
                                    pool.sd = FALSE)
```

```
# Load base R-compatible tools
library(DT)



# Run pairwise t-tests with Holm correction (Welch's method)
pairwise_results <- pairwise.t.test(diet_data$weight_loss,
                                    diet_data$diet.type,
                                    p.adjust.method = "holm",
                                    pool.sd = FALSE)

# Convert to a data frame for display
pval_matrix <- pairwise_results$p.value
pval_df <- as.data.frame(as.table(pval_matrix))
colnames(pval_df) <- c("Group 1", "Group 2", "Holm-adjusted p-value")

# Remove rows with missing values (e.g., B vs B)
```

```
pval_df <- na.omit(pval_df)

# Display DT table (just the body)
datatable(pval_df,
          rownames = FALSE,
          options = list(dom = 't',  # show only table body
                         pageLength = 5))
```

Group 1	Group 2	Holm-adjusted p-value
B	A	0.962241730434801
C	A	0.01934991623641207
C	B	0.01934991623641207

Q5

```
# Sensitivity calculation
sensitivity <- 87 / 108
sensitivity
```

```
[1] 0.8055556
```

```
# Specificity calculation
specificity <- 55 / 64
specificity
```

```
[1] 0.859375
```

```
# Positive Predictive Value
ppv <- 87 / 96
ppv
```

```
[1] 0.90625
```

```
# Negative Predictive Value
npv <- 55 / 76
npv
```

```
[1] 0.7236842
```

$$P(X = 1) = \binom{4}{1} \cdot (0.194)^1 \cdot (1 - 0.194)^3$$

```r
# Sensitivity
sensitivity <- 87 / 108
p_neg <- 1 - sensitivity

# Binomial: exactly 1 of 4 malaria cases test negative
prob_1_neg_out_of_4 <- dbinom(1, size = 4, prob = p_neg)
prob_1_neg_out_of_4
```

```
[1] 0.4065763
```

We are looking for the probability that **2 or more false positives** occur out of 5 people who **do not have malaria**.

This is a **binomial probability** problem with:

- $$n = 5$$

  (number of individuals tested)
- "Success" = a **false positive** test result

From **Part B**, we calculated the test specificity as:

$$\text{Specificity} = 0.859$$

So the probability of a false positive is:

$$P(\text{false positive}) = 1 - \text{Specificity} = 1 - 0.859 = 0.141$$

Computation

We want the probability of **at least 2 false positives**:

$$P(X \geq 2) = 1 - P(X < 2) = 1 - P(X = 0) - P(X = 1)$$

```r
# Define probability of false positive
p_fp <- 1 - 0.859  # = 0.141

# Compute probability of at least 2 false positives in 5 tests
p_at_least_2 <- 1 - pbinom(1, size = 5, prob = p_fp)
p_at_least_2
```

```
[1] 0.1484515
```

The result is approximately **0.148**, meaning there is a **14.8% chance** that at least 2 out of 5 people without malaria would receive a false positive result.